

Neural Networks

GPT-4 has 1.7 trillion parameters. AlphaGo beat Lee Sedol in 2016. ResNet-50 classifies 1,000 categories. All three are made of ONE thing — the perceptron. Today we build it.

How many perceptrons do you think are in a model that can beat the world go champion? Hold that guess — we reveal it at minute 40.

L2

WHAT'S INSIDE

- 01** The perceptron
- 02** The XOR limit — why layers matter
- 03** Activations — pick with intent
- 04** Forward pass — by hand
- 05** A 3-layer MLP in 15 lines

WHY THIS LESSON MATTERS

One lesson. One atom. Every modern AI system.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Describe a perceptron: weights, bias, activation.

2

Explain why a single perceptron cannot solve XOR.

3

Forward-pass a small network by hand.

4

Pick appropriate activations for hidden and output layers.

5

Build a 3-layer MLP in PyTorch in under 20 lines.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

The perceptron

Weighted sum + bias + activation

02

The XOR limit — why layers...

Minsky & Papert 1969 — a 17-year winter

03

Activations — pick with...

ReLU inside; Sigmoid or Softmax at the output. Most networks, most days.

04

Forward pass — by hand

Two inputs, two hidden, one output

01

PART 1 OF 4

The perceptron

Weighted sum + bias + activation

CONCEPT 1 OF 4

The perceptron

Weights — how much each input matters. LEARNED.

Bias — a constant offset. LEARNED.

Activation — introduces non-linearity. CHOSEN.

Output = $\text{activation}(\sum w_i x_i + b)$.

ATOM

1 perceptron

GPT-4 is 1.7 trillion of these. Learn one; understand all.

DEFINITION

The perceptron

Weighted sum + bias + activation

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

The XOR limit — why layers matter

A single perceptron draws one straight line.

XOR requires two lines to separate.

The fix: stack perceptrons with non-linear activations.

Depth unlocks expressiveness.

LESSON

Depth > width

The 1969 AI winter came from misreading a limit of single neurons as a limit of neural networks.

KEY POINTS

- A single perceptron draws one straight line
- XOR requires two lines to separate
- The fix: stack perceptrons with non-linear activations
- Depth unlocks expressiveness

Activations — pick with intent

ReLU inside; Sigmoid or Softmax at the output. Most networks, most days.

DEFINITION

Activations — pick with intent

ReLU inside; Sigmoid or Softmax at the output. Most networks, most days.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Forward pass — by hand

$\text{hidden} = \text{ReLU}(W_1 \cdot x + b_1)$

$\text{output} = \text{Sigmoid}(W_2 \cdot \text{hidden} + b_2)$

$\text{loss} = \text{BinaryCrossEntropy}(\hat{y}, y)$

Code just mechanises what you already did on paper.

DO IT

On paper first

Tiny numeric example before the PyTorch demo.

KEY POINTS

- $\text{hidden} = \text{ReLU}(W_1 \cdot x + b_1)$
- $\text{output} = \text{Sigmoid}(W_2 \cdot \text{hidden} + b_2)$
- $\text{loss} = \text{BinaryCrossEntropy}(\hat{y}, y)$
- Code just mechanises what you already did on paper

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Predict the intermediate shapes

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

- 1
- 2 Predict the intermediate shapes
- 3
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which activation surprised you most?

Q2

Why is "depth > width" a rule of thumb and not a rule?

Q3

What will you type first if I say "build me an MLP"?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

What does a perceptron compute?

A

Average of inputs

B

Weighted sum + bias + activation

C

Max

D

Dot product only

What does a perceptron compute?



CORRECT ANSWER

Weighted sum + bias + activation

WHY

Weighted sum → bias → activation function

Why can't a single perceptron solve XOR?

A

Too few weights

B

Not linearly separable

C

Slow

D

Wrong activation

Why can't a single perceptron solve XOR?



CORRECT ANSWER

Not linearly separable

WHY

XOR requires two lines; a single hyperplane cannot split the four points

Default activation for hidden layers today:

A

Sigmoid

B

Tanh

C

ReLU

D

Softmax

Default activation for hidden layers today:



CORRECT ANSWER

ReLU

WHY

ReLU (and variants) dominate modern hidden layers

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Weights — how much each input matters. LEARNED.

Bias — a constant offset. LEARNED.

Activation — introduces non-linearity. CHOSEN.

Output = activation($\sum w_i x_i + b$).

ATOM

1 perceptron

GPT-4 is 1.7 trillion of these. Learn one; understand all.

THE TAKEAWAY

"Stack perceptrons, add non-linearities, train with data. That is deep learning in one sentence."

WHAT TO NOTICE

1

The perceptron

2

The XOR limit — why layers...

3

Activations — pick with...

4

Forward pass — by hand

Mini assignment — beat the baseline

Train your PyTorch MLP on the breast-cancer dataset from Lesson 1.

DELIVERABLES

- Build the 3-layer MLP from the template.
- Train for 50 epochs with Adam(lr=1e-3).
- Compare test accuracy to the LogisticRegression baseline.
- Post both numbers + one observation on the cohort forum.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Notebook URL with training curve + comparison.

ONE THING TO REMEMBER

“

*"Stack perceptrons, add non-linearities, train with data.
That is deep learning in one sentence."*

— AI Learning Hub

END OF LESSON 3

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
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NEXT → Lesson 3 — Backpropagation. How those weights actually learn.